INTRODUCTION TO C++

BASICS OF C++

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Types of Programming Language

IDE (Integrated Development Environment)

Basic Structure of a C++ Program

Identifiers

Fundamental Data Types

Compound Data Types

TYPES OF PROGRAMMING LANGUAGE

Machine Language

The natural language that the computer understands (**0's** and **1's**)

Sample program to add 5 and 3

```
10111000
00000101
00000000
10111011
00000011
00000000
00010001
11011000
```

Assembly Language

A programming language that uses "**abbreviations**" or **mnemonics** in place of binary patterns

Sample program to add 5 and 3

```
MOV AX, 05
MOV BX, 03
ADD AX, BX
```

Assembler

A special program that **translates assembly** language mnemonics into machine language

Sample program to add 5 and 3

```
x = 5;
y = 3;
sum = x + y
```

High-Level Language

A programming language that uses **english-like** commands or instructions

Compiler

A special program that **translates high-level** language instructions into machine language

HIGH-LEVEL PROGRAMMING LANGUAGES

python™

Java™

SQL

C++

HIGH-LEVEL PROGRAMMING LANGUAGES

python™

Java™

SQL

C++

IDE

Integrated Development Environment (IDE)

A software application that provides comprehensive facilities to programmers for software development

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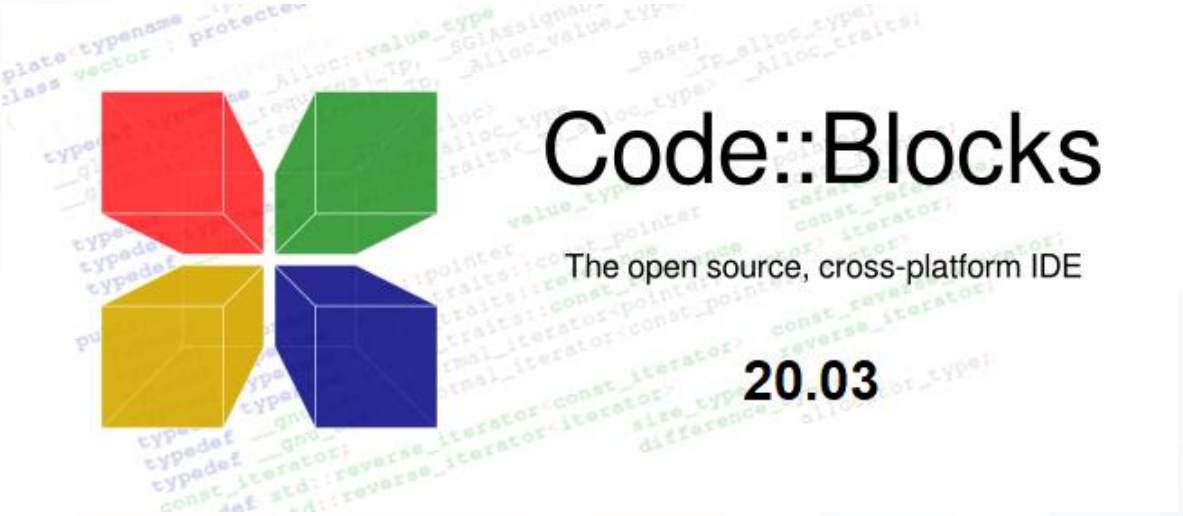
Integrated Development Environment (IDE)



Consists of a **source code editor**, build automation tools, and debugger

BASIC STRUCTURE OF A C++ PROGRAM

STRUCTURE OF C++ PROGRAM

```
// My First C++ Program

#include <iostream>

using namespace std;

int main ( )

 {

     cout << “Hello World!”;

     return 0;

}
```

```
// My First C++ Program

#include <iostream>

using namespace std;

int main ( )

 {

     cout << "Hello World!";

     return 0;

}
```

Comments do not affect the operation of the program; however, they provide an important tool to **document** directly the source code what the program does and how it operates

// line comment

/* block comment */

HEADER IOSTREAM

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
 {
    cout << "Hello World!";
    return 0;
}
```

Instructs the preprocessor to include a section of standard C++ code, known as header iostream, that allows to perform **input** and **output** operations

```
// My First C++ Program

#include <iostream>

using namespace std;

int main ( )

 {

     cout << "Hello World!";

     return 0;

}
```

GWENCHANA

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
 {
    cout << "Hello World!";
    return 0;
}
```

All the elements in the standard C++ library are declared within what is called a **namespace**: the namespace std

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
 {
    cout << "Hello World!";
    return 0;
}
```

A function is a group of **code statements** which are given a name: in this case, this gives the name "main" to the group of code statements that follow

FUNCTION'S BODY

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
{
    cout << "Hello World!";
    return 0;
}
```

{ } , everything between these braces is the function's body that defines what happens when "main" is called

INSERTION OPERATOR

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
 {
    cout << "Hello World!";
    return 0;
}
```

<< , indicates that what follows is inserted into "cout"

STATEMENT

```
// My First C++ Program
#include <iostream>
using namespace std;
int main ( )
 {
    cout << "Hello World!" ;
    return 0 ;
}
```

One of the most common syntax errors in C++ is forgetting to end a statement with a **semicolon**

IDENTIFIERS

IDENTIFIERS

A **valid identifier** is a sequence of one or more letters, digits, or underscore (_).

number

age_in_year

result_1

The C++ language is a "**case sensitive** language"

The **Result** variable is **not** the same as the **result** variable

Reserved keywords cannot be use as identifiers

refer to C++ documentation for the complete list

class

auto

catch

final

CODING EXERCISE

```
#include <iostream>
using namespace std;
int main( ) {
    int age = 18;
    cout << "My age is";
    cout << age;
    return 0;
}
```

FUNDAMENTAL DATA TYPES

CHARACTER

The most basic type is **char**, which is a **one-byte** character

Represent a **single** character, such as ‘A’ or ‘$’

```
#include <iostream>
using namespace std;
int main( ) {
    char middle = ‘A’;
    ----------------------
    ----------------------
}
```

Store a **whole number** value, such as 7 or 1024

Can either be **signed** or **unsigned**

```
#include <iostream>
using namespace std;
int main( ) {
    int age = 18;
    ----------------------
    ----------------------
}
```

FLOATING POINT

Represent **real values**, such as 3.14 or 0.01

```
#include <iostream>
using namespace std;
int main( ) {
    float weight = 51.8;
    ----------------------
    ----------------------
}
```

BOOLEAN

Can only represent one of two states, **true** or **false**

```
#include <iostream>
using namespace std;
int main( ) {
    bool result = true;
    ----------------------
    ----------------------
}
```

COMPOUND DATA TYPES

STRING CLASS

Store **sequences** of **characters**, such as words or sentences.

```
#include <iostream>
#include <string>
using namespace std;
int main( ) {
    string name = "Gyro";
    string last_name = "Madrona";
    cout << "Hello, My first name is ";
    cout << name << endl;
    cout << "My last name is ";
    cout << last_name;
    ------------------------------------
    ------------------------------------
```

LABORATORY